Introduction to Prolog Language

CS 3111 Practical Work on CS 3121

Logic Programming and Expert System

What is Prolog?

- Prolog stands for PROgramming in LOGic.
- It is a logic programming language based on formal logic, particularly first-order predicate logic. Unlike most traditional languages that are procedural (focus on *how* to do things), Prolog is declarative (focuses on *what* is true).
- It is one major example of the fourth-generation language that supports the declarative programming paradigm.
- This is particularly suitable for programs that involve symbolic or non-numeric computation. This is the main reason to use Prolog as the programming language in artificial intelligence, where symbol manipulation and inference manipulation are the fundamental tasks.

What is Procedural Programming?

- In **procedural programming**, you write a sequence of instructions to tell the computer what to do step by step.
- You explicitly state how to compute addition and substraction.

```
public static int add(int x, int y) {
    return x + y;
}
public static int subtract(int x, int y) {
    return x - y;
}
```

```
public static void main(String[] args) {
    int a = 10;
    int b = 5;

    System.out.println("Addition: " + add(a, b));
    System.out.println("Subtraction: " + subtract(a, b));
    System.out.println("Multiplication: " + multiply(a, b));
    System.out.println("Division: " + divide(a, b));
}
```

What is Declarative Programming?

- **Declarative programming** describes *what* the program should accomplish rather than *how* to accomplish it.
  - You declare **facts** and **rules**.
  - The system uses **logical inference** to answer queries.
- Example -

  father(john, mary).

  father(john, david).

  - This means "John is the father of Mary and David". You don't tell Prolog *how* to search or *what* to do procedurally — just what is true.

```
SELECT name FROM students WHERE grade = 'A';
"Give me the names of all students who have an A grade."

You don't specify how to search, loop, or filter — the
database engine figures that out for you.
```

Prolog vs Procedural Language.

Feature	Procedural Language (e.g. C, Python)	Prolog
Paradigm	Imperative / procedural	Declarative / logic-based
Control	Programmer controls flow	System decides logic and flow
Focus	How to solve (algorithms)	What is true (relationships)
Program Structure	Functions, loops	Facts, rules, queries
Example	Write loops, recursion	Define rules, let Prolog solve

Applications of Prolog.

- Expert systems - An **Expert System** is a **computer program** designed to **simulate the decision-making ability of a human expert** in a specific domain (like medicine, agriculture, engineering, etc.).
- Natural Language Processing (NLP) - **Natural Language Processing (NLP)** is the field of AI that enables computers to **understand**, **interpret**, and **generate** human language (like English, Sinhala, Tamil, etc.).
- AI and knowledge representation - **Knowledge Representation (KR)** is about **storing information** in a way that a computer can **reason** about it — like facts, rules, and relationships.
- Rule-based reasoning systems - A **rule-based system** is an AI that uses **IF-THEN rules** to make decisions or draw conclusions.

Environment Setup

- **Prolog Compiler / Interpreter**

A **Prolog** program can be compiled to native code to give rise to a machine-dependent executable using the **Prolog compiler** or the **Interpreter**.

Example –

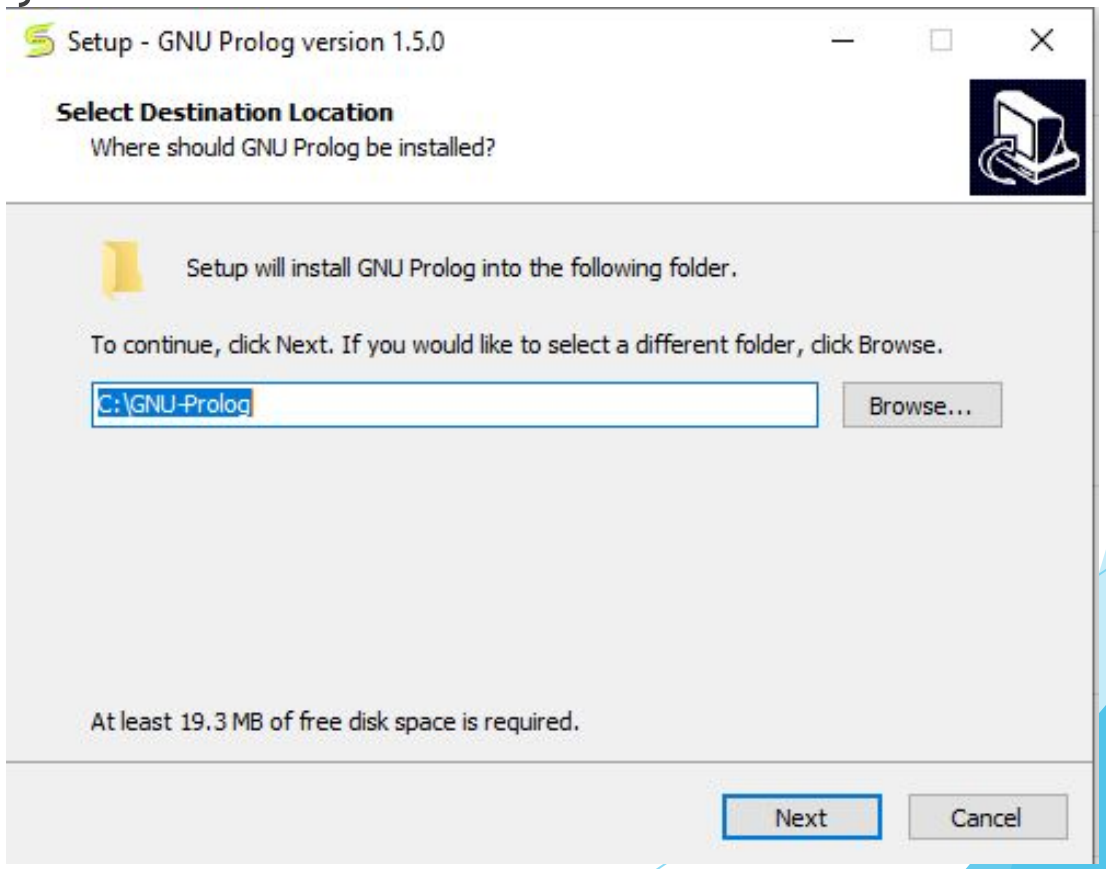
1. GNU Prolog
2. SWI-Prolog etc.

Environment Setup

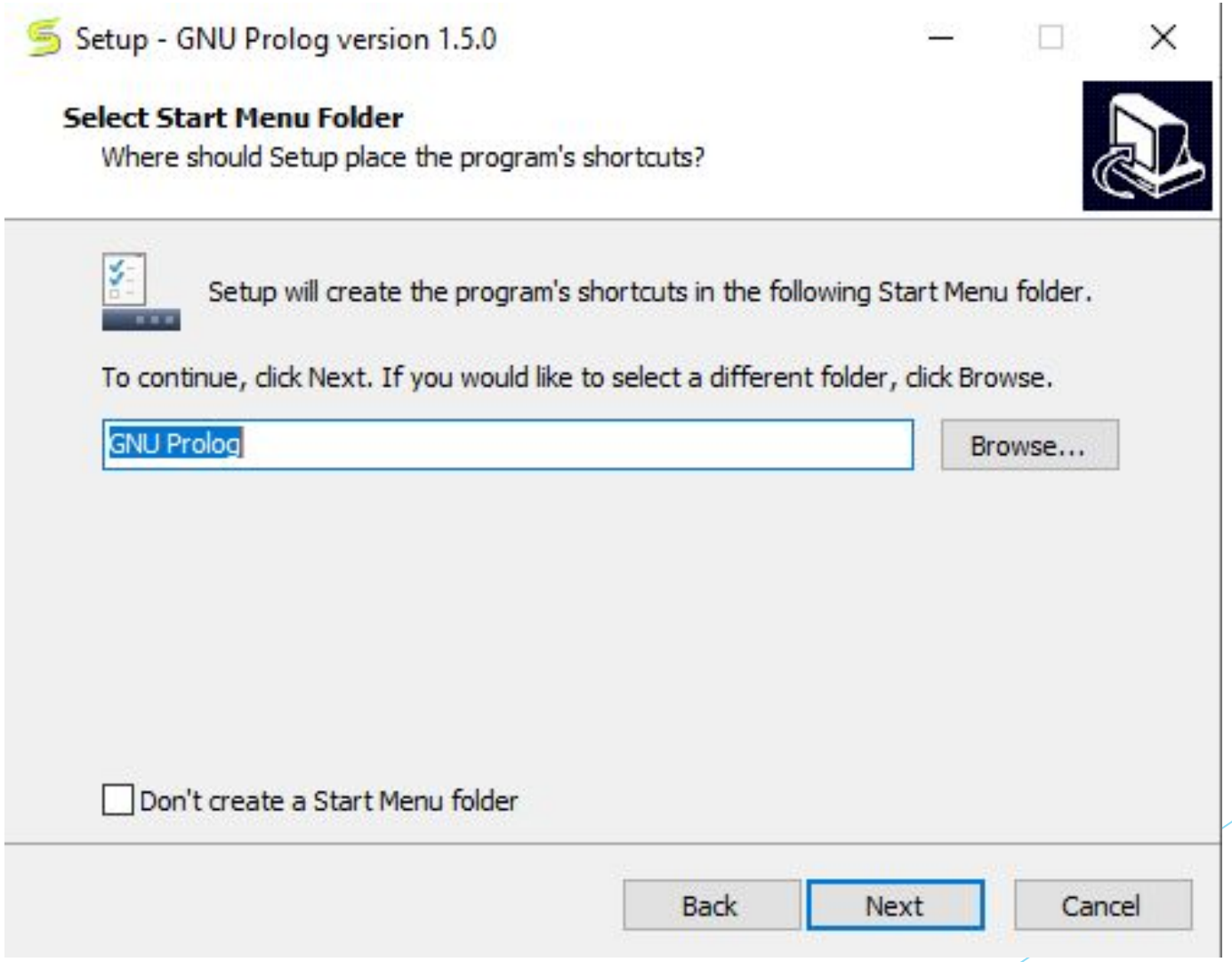
- Download GNU Prolog from the following link.

  http://www.gprolog.org/

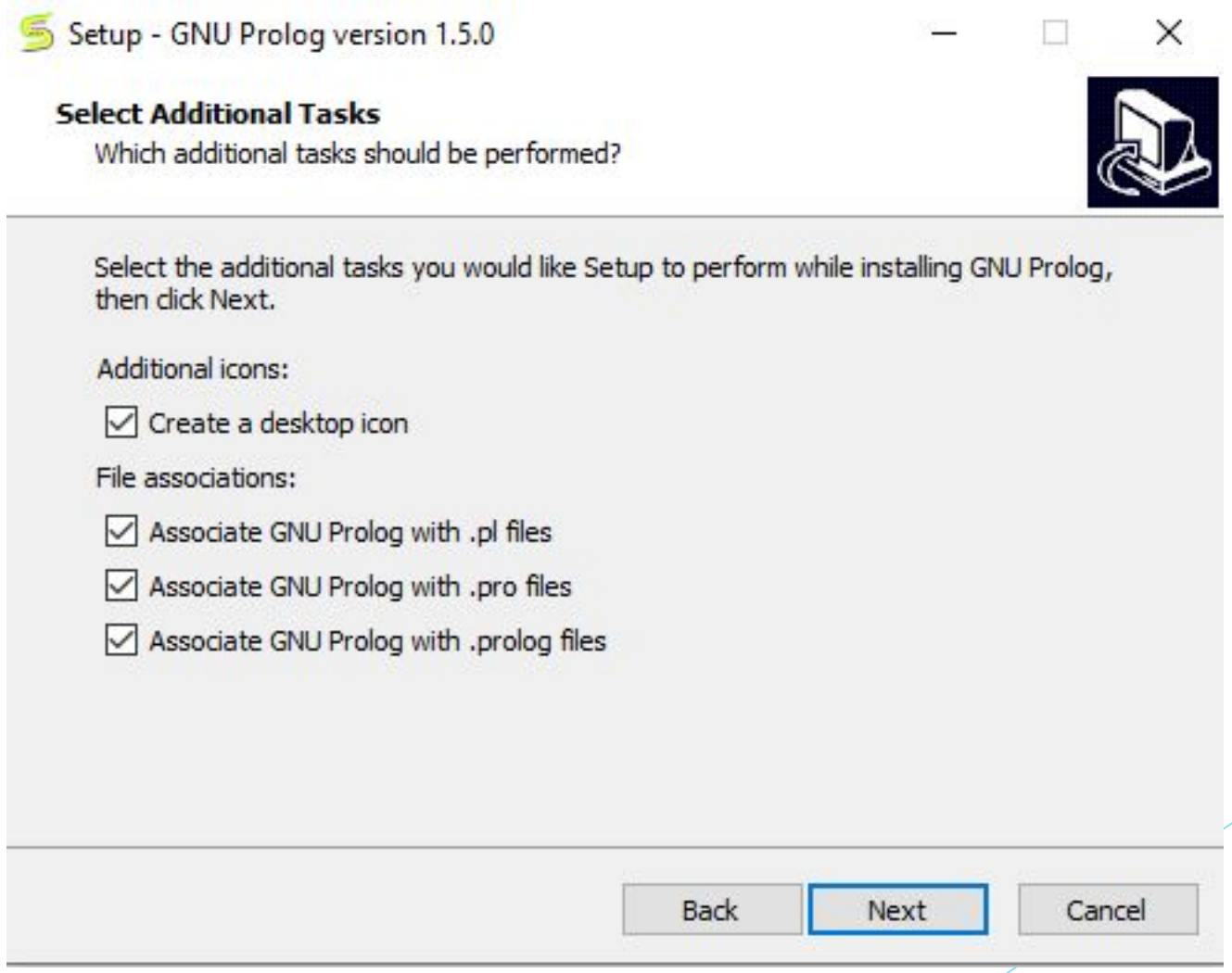
- Download the exe file and run it.
- Select proper directory where you want to install the software, otherwise let it be installed on the default directory. Then click on next.



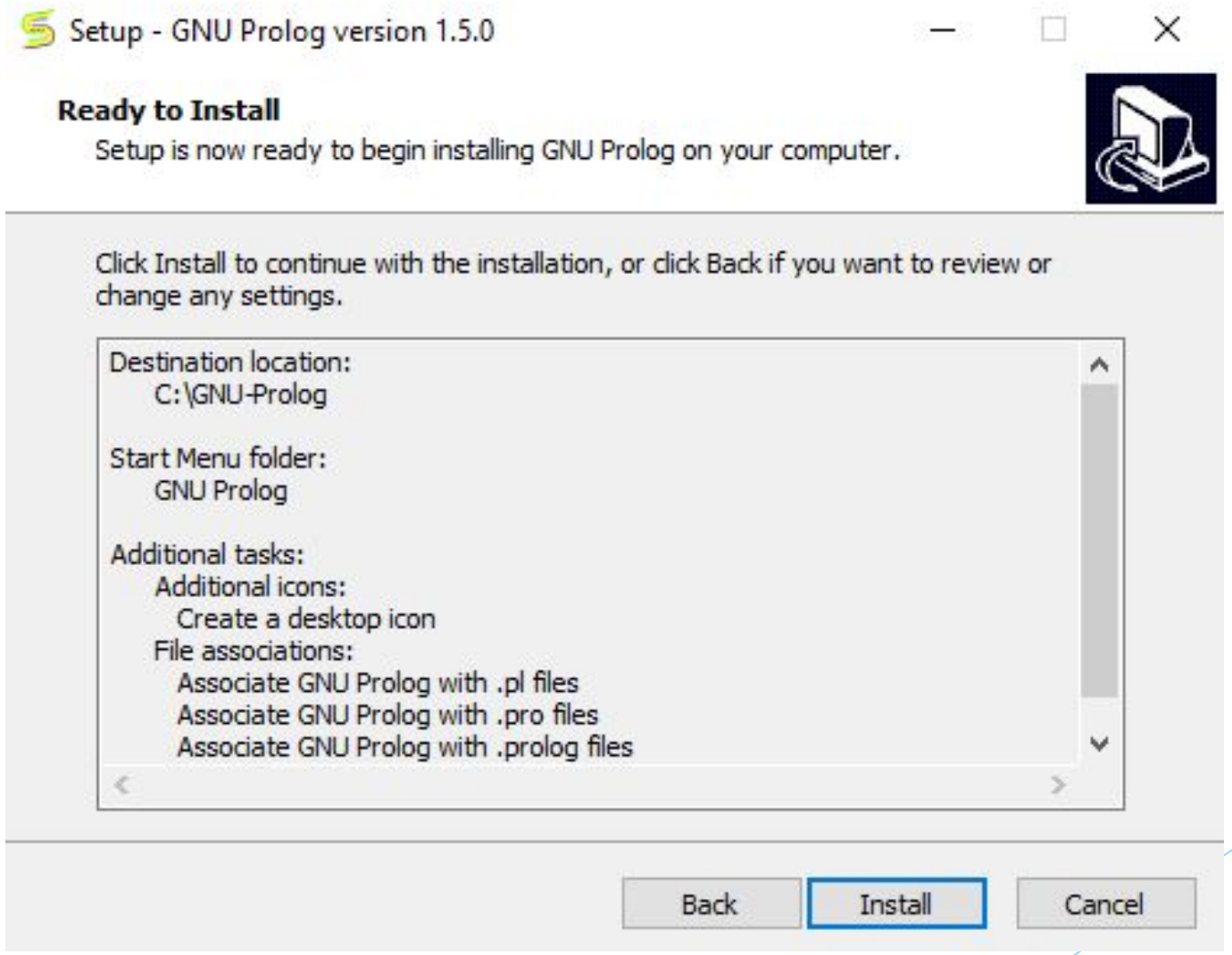
- You will get the below screen, simply go to next.



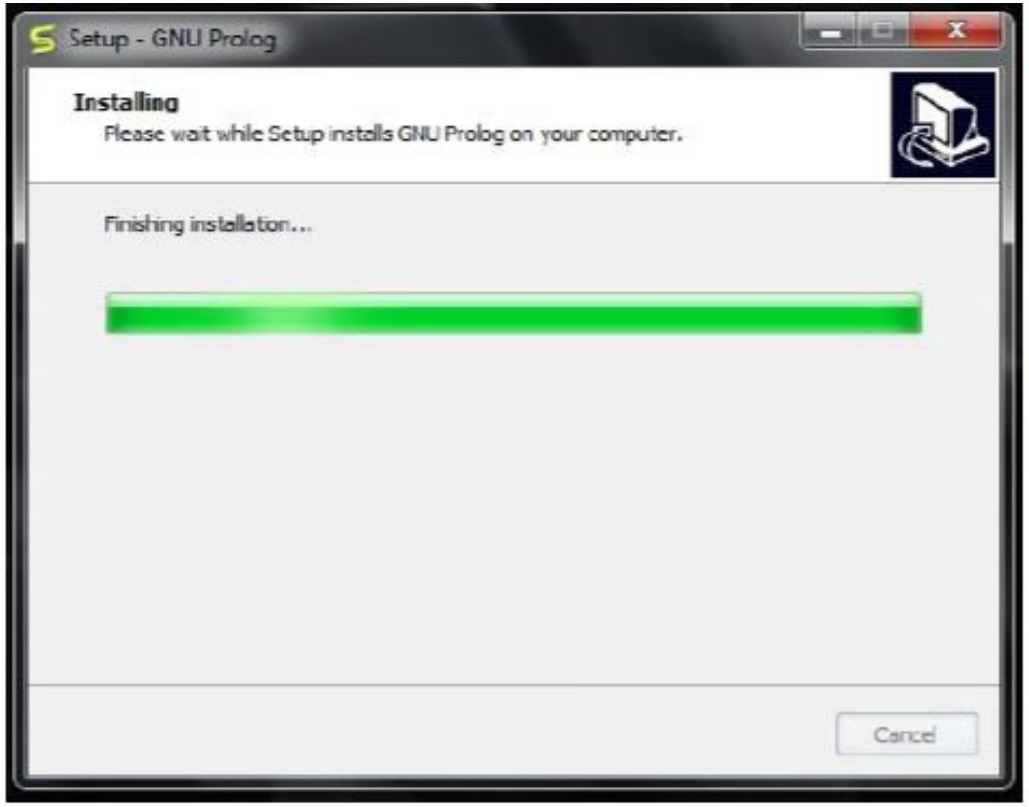
- You can verify the below screen, and check/uncheck appropriate boxes, otherwise you can leave it as default. Then click on next.



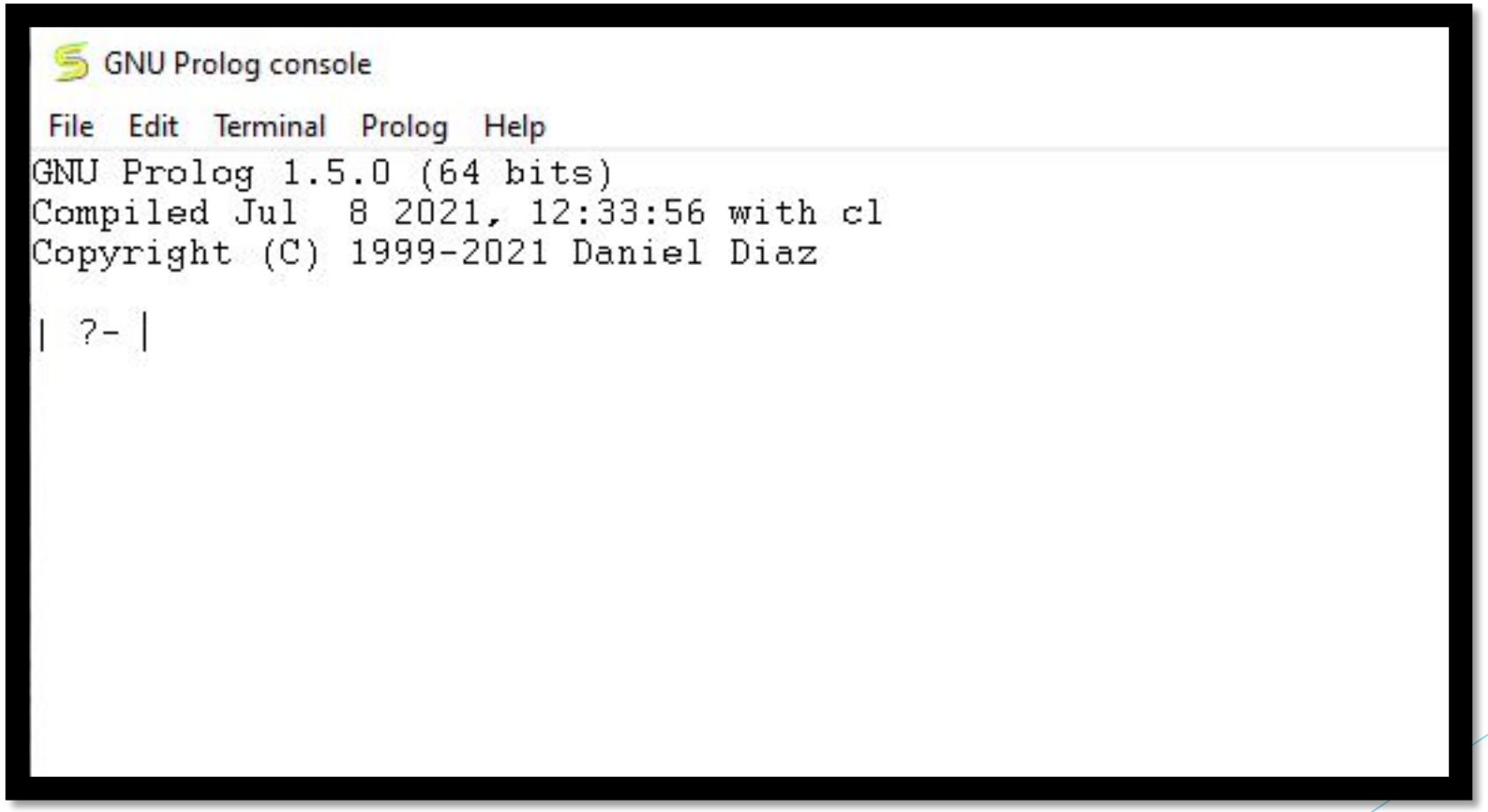
- In the next step, you will see the below screen, then click on Install.



- Then wait for the installation process to finish and finally click on Finish to start GNU Prolog.



- The GNU prolog is installed successfully as shown below.



Testing your code in GNU Prolog.

- In **Prolog**, built-in functions (called **built-in predicates**) are provided by the language to perform common tasks such as I/O, arithmetic, list handling, control flow, and more.
- Open GNU prolog and do the following at the prompt.

```
| ?- write('Hello World').
Hello World
```

- Write a program to print 'Hello' on the first line and 'World' on the second line.

```
| ?- write('Hello'),nl,write('World').
Hello
World
```

- ‘Write’ is a built in function in prolog.
- To **add two numbers in Prolog**, you can use the built-in arithmetic evaluation with Is/2.

```
| ?- X is 3+8.

X = 11
```

Another way is

- Open Notepad ++ and type the following.

```
hello :-
    write('Hello World').
```

- Save this as **first.pl** file in the directory on your desktop.
- Locate this directory using GNU prolog using File >Change Dir… in GNU Prolog.
- Load the file like the following in GNU Prompt.

```
yes
| ?- [first].
```

- Your code is compiled now by GNU Prolog.
- Now type this

```
yes
| ?- hello.
Hello World
```

- You can see the output in GNU Prolog.

```
| ?- change_directory('C:/Users/Trio/Desktop/testcode').

yes
| ?- [first].
compiling C:/Users/Trio/Desktop/testcode/first.pl for byte code...
C:/Users/Trio/Desktop/testcode/first.pl compiled, 2 lines read - 293 bytes written, 6 ms

yes
| ?- hello.
Hello World
```

Thank You!